When Should an AI Tutor Abstain?
Phase 1: State of the Art, Datasets and Solution Concept (Topic 15)

Matyáš Veselý
album K-9444
01219835@pw.edu.pl

Grzegorz Kacprowicz
album 286358
01124547@pw.edu.pl

Wiktoria Koniecko
album 320633
01171239@pw.edu.pl

Faculty of Mathematics and Information Science, Warsaw University of Technology
supervisor: Anna Wróblewska, anna.wroblewska1@pw.edu.pl

Abstract

We study a tutor that answers NLP course questions from a fixed collection of course documents and abstains when the retrieved documents lack adequate evidence. Before the answer is generated, a gate scores whether the retrieved passages are sufficient; if not, the tutor declines and names what is missing. We compare three such gates (retrieval-score statistics, a small extractive reader and the language model judging its own evidence) against ordinary retrieval-augmented generation (RAG), using the same retriever and generator, at matched coverage. Deliverables: a 120-question course benchmark with human-checked evidence, and risk–coverage results for all systems.

1 State of the art

1.1 Foundations of retrieval and selective answering

Lewis et al. (2020) combine a dense Wikipedia retriever with a sequence-to-sequence generator. On Natural Questions, TriviaQA, WebQuestions and CuratedTREC, RAG improves open-domain question answering (QA) over strong parametric or extractive baselines. However, answer correctness does not establish support in the retrieved evidence. Rajpurkar et al. (2018) introduce SQuAD 2.0, adding adversarial, human-written unanswerable questions to Wikipedia paragraphs. Readers perform substantially below humans (Table 1), establishing a useful model for constructing plausible unsupported questions.

Geifman and El-Yaniv (2017) attach a rejection rule to neural classifiers and evaluate on CIFAR-10, CIFAR-100 and ImageNet. Their experiments show that rejecting uncertain cases reduces error while retaining substantial coverage, with statistical risk control under their assumptions. Kamath et al. (2020) train a correctness calibrator on in-domain and known out-of-domain QA. On mixtures of SQuAD and unseen QA domains, it answers more questions at a fixed accuracy than maximum model probability. This supports reporting the complete risk–coverage curve.

1.2 Evidence sufficiency and adaptive retrieval

Slobodkin et al. (2023) probe hidden states and vary prompts on SQuAD 2.0, Natural Questions and MuSiQue. Joren et al. (2025) define sufficient context and validate an automatic judge on a human-labelled mixture of PopQA, FreshQA, Natural Questions and EntityQuestions. Their judge reliably detects sufficiency; selective generation on HotpotQA and MuSiQue improves correctness among answered questions. This is the closest conceptual precedent for our gate.

Jeong et al. (2024) use a question-complexity classifier to route SQuAD, Natural Questions, TriviaQA, MuSiQue, HotpotQA and 2WikiMultiHopQA queries to no, single-step or iterative retrieval. Adaptive-RAG improves the accuracy–cost trade-off, motivating a query-only control rather than an evidence-sufficiency claim. Moskvoretskii et al. (2025) compare adaptive retrieval pipelines and uncertainty estimators on test subsets of the same six datasets.

Asai et al. (2024) train Self-RAG to retrieve and critique relevance, support and utility. Experiments on PopQA, TriviaQA, PubHealth, ARC, biographies and ASQA show improved QA and citation quality. Its trained generation-time critic is a stronger architectural intervention than a standalone gate. Veselý et al. (2026) predict retrieval sufficiency from score-distribution features on ViDoRe and a private Czech regulatory corpus. GeneralQPP outperforms classic predictors and a local multimodal judge on ViDoRe.

1.3 Faithfulness evaluation and educational systems

Ming et al. (2025) introduce FaithEval with unanswerable, conflicting and counterfactual contexts derived from established QA datasets. Even strong models frequently fail its unanswerable-context task, motivating explicit missing-evidence evaluation. Gao et al. (2023) introduce ALCE, evaluating correctness and citation support on ASQA, QAMPARI and ELI5. Response reranking improves citation quality, while substantial support gaps remain.

Niu et al. (2024) annotate hallucinated spans in RAGTruth responses for QA, summarization and data-to-text generation. A fine-tuned detector improves response selection and reduces hallucinations; the reported benefit depends on how many responses are retained. Fernandez et al. (2024) introduce SyllabusQA, course-logistics questions over real syllabi, including insufficient-information cases. Fine-tuning plus retrieval improves factuality, but remains below the separately evaluated human reference. Ostrowska et al. (2026) integrate a grounded tutor into Moodle and evaluate internal STEM and humanities materials and user feedback.

Project gap. These works justify a controlled comparison of score-based, small-reader and language-model sufficiency gates on a fixed course corpus. The contribution is this comparison under answerable, partial and missing evidence, with human-checked support at matched coverage; it is not the invention of abstention or sufficient-context judging.

Published quantitative evidence. Each row of Table 1 identifies the evaluation data and split. Comparisons are within rows; different metrics, models and datasets do not form a common leaderboard. Percentages and fractions retain the papers' scales. References point to published versions.

2 Proposed approach

All systems share one retrieval stage and differ only after it (Figure 1). We consider two retrieval configurations. In the textual configuration, the 5 to 10 authorised course files are split into passages of at most 256 tokens with a 32-token overlap and indexed once with BM25 on CPU. In the visual configuration, the retrieval unit is the page: page images are indexed once with a vision-language late-interaction retriever (Faysse et al., 2025), the generator reads the retrieved pages as images (Qwen2.5-VL-3B-Instruct, the vision-language model of the same family), and the components that need text (B1, G2) use the text layer of those pages. In both configurations every question retrieves the same top-5 units, called passages below. The gated system computes a sufficiency score before generation. If the score is greater than threshold τ, the plain-RAG prompt runs unchanged; otherwise the system abstains and names the missing answer point. A missing-source stress test deletes the gold source file from the index and reruns everything.

Systems compared. Table 2 lists the seven arms: two baselines, a confidence-thresholded variant, a query-only control and the three sufficiency gates.

For each scorer, the threshold τ is set on the validation split so that coverage equals that of B2, then swept for the full risk–coverage curve. The gated models are compared with the RAG baseline at matched coverage, Bonferroni-corrected over three arms. The ranking among scorers is descriptive. LLM systems run three seeds at temperature 0.3; we report the mean, standard deviation and 1,000-sample bootstrap intervals over questions. With 80 test items, differences below roughly 20 percentage points in the unsupported-answer rate are not detectable at the corrected significance level and are reported as undetectable, not absent. An error analysis groups at least 20 failures into named types.

Metrics.

- **unsupported-answer rate at matched coverage** (primary metric) — the share of answered questions not entailed by the cited passages;
- **coverage** and **area under the risk–coverage curve** (AURC) over the threshold sweep;
- **answerability F1** with correct and false refusals;
- **gate latency** and **GPU memory** (VRAM), to quantify the cost of each gate.

3 Datasets and their status

We work with one dataset that we build ourselves and potentially a public one (Table 3). The first one is marked TO COLLECT (it does not exist yet and we create it) and the second is marked READY (public and usable now). The main experiment runs on our own course question set. ViDoRe V3 is our fallback should the course materials not be available, and otherwise a potential second, open test set on which we check whether our findings apply beyond one course.

Table 1: Published quantitative evidence. Comparisons are valid within a row only. *FaithEval source datasets: BioASQ, DROP, HotpotQA, Natural Questions, NewsQA, RACE, SQuAD, SearchQA, TextbookQA and TriviaQA. QPP: query performance prediction. DPR: Dense Passage Retrieval. RAGAS: Retrieval-Augmented Generation Assessment. OOD: out of domain.

Work / metric	Dataset and evaluation split	Published result and interpretation
RAG (Lewis et al., 2020) Exact match	Natural Questions open-domain test, using the paper's QA protocol; Table 1.	RAG-Sequence 44.5 %; DPR baseline 41.5 %. Retrieval improves correctness; this is not a faithfulness or abstention score.
SQuAD 2.0 (Rajpurkar et al., 2018) Answer F1	SQuAD 2.0 (SQuADRun), official test; Table 3.	DocQA+ELMo 66.3; human 89.5. Both answerable and unanswerable items contribute to this metric.
Selective prediction (Geifman and El-Yaniv, 2017) Risk / coverage	CIFAR-10, reserved half of the validation set (about 5,000 images), used as test; Table 1.	At target risk 0.01, observed test error 0.0092 and coverage 0.7856. Vision result; not a tutor risk guarantee.
Selective QA (Kamath et al., 2020) Coverage	Paper-defined test: 4,000 SQuAD + 4,000 unseen-domain items, sampled from disjoint MRQA validation data. Mean over 20 ordered domain pairings; Table 1.	At 80 % exact-match accuracy, calibrator answers 56.1 %, versus MaxProb 48.2 %. OOD domains: NewsQA, TriviaQA, SearchQA, HotpotQA and Natural Questions.
Unanswerability (Slobodkin et al., 2023) Classification F1	Paper-defined test constructed from SQuAD 2.0 development, 11,873 items; Table 2(a), zero-shot.	Flan-UL2: 46.3 with ordinary prompt, 92.3 with unanswerability hint. This is classification F1, not answer-token F1.
Sufficient Context (Joren et al., 2025) Accuracy; F1	Dedicated 115-item evaluation set, sampled from PopQA, FreshQA, Natural Questions and EntityQuestions; Table 1. Demonstration held out.	Gemini 1.5 Pro, one-shot: accuracy 0.930, F1 0.935; zero-shot: 0.870 and 0.878. Evaluates the judge, not end-to-end tutor risk.
Adaptive-RAG (Jeong et al., 2024) Answer F1	Natural Questions, paper's 500-query evaluation/test subset; FLAN-T5-XL; Table 2.	Adaptive-RAG 47.30; iterative retrieval 47.80; single-step retrieval 47.30. Adaptive routing does not always improve accuracy.
Uncertainty routing (Moskvoretskii et al., 2025) In-Accuracy (answer inclusion)	Natural Questions, 500-query test subset; Llama-3.1-8B; Table 1.	Lexical-similarity uncertainty: 0.512, with 0.58 retrieval calls/query; always retrieve: 0.496, with 1.00 call/query.
Self-RAG (Asai et al., 2024) Answer inclusion	PopQA long-tail evaluation subset, 1,399 rare-entity questions; Table 2.	Self-RAG 13B: 55.8 %; retrieval-augmented Llama2-chat 13B: 51.8 %. Gold-answer inclusion is less strict than exact match.
GeneralQPP (Veselý et al., 2026) AUROC	ViDoRe, eight domains; held-out 20 % per domain from stratified 80/20 splits, seed 42; weighted mean, Hit@10 target.	GeneralQPP 0.856; Classic Full 0.835; local multimodal judge 0.649. Predicts fully relevant evidence in top retrieved pages, not final-answer support.
FaithEval (Ming et al., 2025) Strict accuracy	FaithEval Unanswerable Context evaluation set; Table 4, AVG across ten source datasets.*	GPT-4o 0.597; Claude 3.5 Sonnet 0.621. Success means recognizing that the provided context cannot answer the question.
ALCE (Gao et al., 2023) Citation recall / precision	ALCE-ASQA evaluation sample of 1,000 ASQA development questions; Table 4.	ChatGPT, five-passage prompt: 73.6 / 72.5; with response reranking: 84.8 / 81.6. Reranking uses generated candidates, unlike a pre-generation gate.
RAGTruth (Niu et al., 2024) Hallucination rate	RAGTruth 450-instance test, 150 per task (QA, summarization, data-to-text); Table 7, Llama-2-7B/Mistral-7B response pair.	Random selection: 52.4 % over 450 responses; select fewer detected spans: 41.1 % over 450; accept only zero detected spans: 19.3 % over 328. The last comparison changes coverage.
SyllabusQA (Fernandez et al., 2024) Fact-QA F1	Model: SyllabusQA test; human: 20 % subset of test; syllabi disjoint across train/validation/test; Table 4.	LLaMA-2-13B + supervised fine-tuning + RAG + chain of thought: 0.597; human 0.685. The human figure is not on the full model-evaluation split.
Moodle tutor (Ostrowska et al., 2026) RAGAS faithfulness	Internal STEM and humanities evaluation; dataset identity, size and train/test split not reported in the demo paper; Section 5.	STEM 0.97; humanities 0.98. Automatic judge scores; not a public held-out benchmark or an abstention result.

Table 2: Systems compared. All arms consume the identical top-5 retrieved units (BM25 passages or ColPali pages); they differ only in the decision taken before (or instead of) generation.

Arm	Decision rule	Cost of the decision
B1 classical baseline	Highest-scoring sentence of the top passage; abstains if the top retrieval score is below τ_1 (tuned on validation)	microseconds
B2 plain RAG (control)	Qwen2.5-3B-Instruct (textual) or Qwen2.5-VL-3B-Instruct (visual), 4-bit, five passages; answers with citations or says the materials do not contain the answer	none
B2+conf	B2 plus post-hoc verbalised confidence, thresholded	one extra LLM call
G0 query-only control	Prompt of G3 without the passages; a gate that does not beat G0 is not reading the evidence	one LLM call
G1 score-only	Training-free retrieval-score statistics (BM25 or late-interaction scores): top score s_1, relative gap $(s_1 - s_2)/s_1$, number of top-5 scores above $s_1/2$; equal-weight rank average	microseconds
G2 small reader	Released SQuAD 2.0 extractive reader (DeBERTa-v3-base, 184M parameters); s = largest best-span minus no-answer margin	milliseconds, CPU
G3 LLM self-judge	The same Qwen model rates whether the passages answer q fully, partly or not; $s = P(\text{fully}) + \frac{1}{2}P(\text{partly})$	seconds, one LLM call

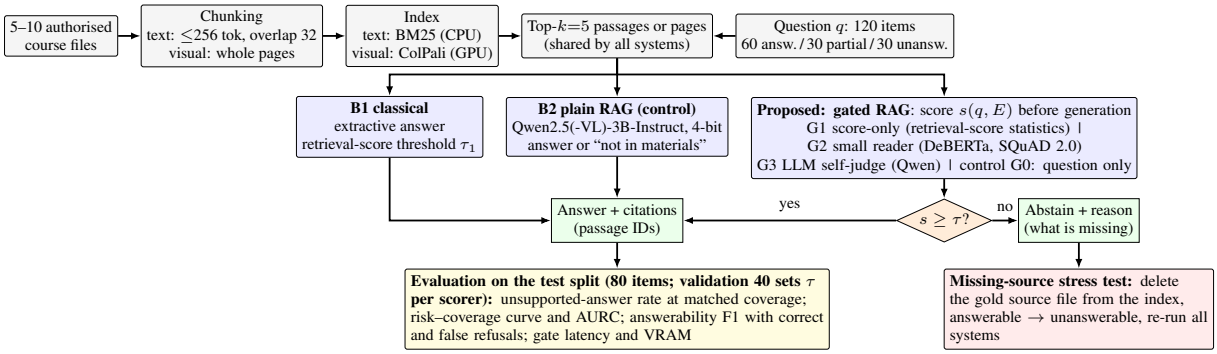


Figure 1: Pipeline: shared retrieval in a textual (256-token passages, BM25) or visual (pages, ColPali) configuration, three system families and one gate decision before the generator.

Table 3: The two datasets the project touches, with their status, size, label definition, class balance, licence and download route.

	1. Course question set (our benchmark)	**2. ViDoRe V3 (Loison et al., 2026)**
Status	**TO COLLECT**	**READY**
Size	120 questions over 5 to 10 course files on one NLP topic; 40 for validation and 80 for testing.	Ten corpora of visually rich documents from professional domains (finance, computer-science textbooks, pharmaceuticals, human resources, industrial maintenance, physics lectures, energy): about 26,000 pages and 3,099 human-verified queries. Eight corpora are public (19,256 pages, 2,419 queries); two are private hold-out sets. Documents are in English or French; every query is available in English, French, German, Italian, Portuguese and Spanish.
Label definition	Every question is labelled answerable, partially answerable (one of its answer points is missing from the materials) or unanswerable (on topic and plausible, but no passage supports an answer). For every question we also list the expected answer points and the passages that contain them.	Each query comes with its relevant pages, bounding boxes around the evidence, a verified reference answer, and a query type (open-ended, extractive, numerical, multi-hop, compare-contrast, boolean or enumerative).
Class balance	60 answerable / 30 partially answerable / 30 unanswerable (50 % / 25 % / 25 %).	In the original release every query is answerable from the corpus. We obtain our three classes from the annotated evidence: we select 120 queries across the public corpora and, for 30 of them, remove all relevant pages from the index (unanswerable); for 30 queries with two or more relevant pages, we remove some of them (partially answerable); the other 60 stay unchanged. This reproduces the 60 / 30 / 30 balance of dataset 1.
Licence	The course files remain the lecturer's property; we use them with her permission and do not redistribute them. Our questions and labels: CC BY 4.0, or internal use only if the lecturer prefers.	Annotations, relevance judgments and metadata: CC BY 4.0 (per the Hugging Face dataset cards). The documents were selected from openly licensed governmental, educational and enterprise sources; each carries its own licence in the metadata.
How obtained	The course files are requested from the lecturer. The questions are written by the team from these files, following the SQuAD 2.0 guideline for unanswerable questions.	Hugging Face Hub, collection `vidore/vidore-benchmark-v3` (eight public datasets `vidore/vidore_v3_*`); we use the page images in the visual configuration and the text layer of the pages in the textual one. The removal step is scripted, and Grzegorz and Wiktoria check every modified item.
How many, by when	All 120 questions (60 / 30 / 30) by 1 December 2026 (week 4), before the Phase 2 deadline on 2 December.	Complete and public already; the 120-query subset is derived in week 1.
Annotated by whom	Grzegorz Kacprowicz and Wiktoria Koniecko label every question independently. We report agreement as Cohen's κ for the class label and as span F1 for the evidence passages. Disagreements are resolved by a third annotator, Matyáš Veselý.	Annotated by the benchmark's authors; Grzegorz and Wiktoria check every item we modify.
Role	The main experiment runs on this set.	Plan B: if the course files are not available, this benchmark replaces dataset 1. Otherwise it is a potential second test set on open data.

4 Success deliverables

D1. The 120-question course benchmark with class labels, answer points, evidence passages and inter-annotator agreement (Cohen's κ or span F1).
D2. Risk–coverage curves and matched-coverage unsupported-answer rates for B1, B2, B2+conf and G0–G3 on the 80-item test split, with bootstrap intervals, the blind human check of 60 answered items per system and the missing-source stress test.
D3. An error taxonomy of at least 20 named failure types, together with code, prompts and configurations that rerun every experiment on CPU plus one consumer GPU.

5 Week-by-week plan

Table A1 (Appendix A) assigns every week to named owners (MV: Matyáš Veselý; GK: Grzegorz Kacprowicz; WK: Wiktoria Koniecko). Course deadlines are set in bold.

Data statement

Source. 5 to 10 lecture files of one NLP course, supplied by the lecturer; fallback: the eight public ViDoRe V3 corpora from the Hugging Face Hub.
Licence. The course files remain the lecturer's

property and are not redistributed; ViDoRe V3 annotations are CC BY 4.0 and each of its documents carries its own open licence in the metadata.

Personal data. The course files are teaching materials and contain no student data; before indexing we scan them for e-mail addresses, names of third parties and other personal identifiers and remove any found. The questions are written by the team and contain no personal data.

Annotation. Performed by named team members, not by crowd workers; agreement statistics are reported.

Processing. The retriever, generator, reader and gates run locally; the course files are not uploaded to external model providers. If an external LLM judge is used, it receives only retrieved passages and generated answers, and this is stated in the final report.

Release. The published benchmark contains questions, labels, passage identifiers and character offsets, not the course files themselves.

References

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *Proc. ICLR 2024*.

Manuel Faysse, Hugues Sibille, Tony Wu, Bilel Omrani, Gautier Viaud, Céline Hudelot, and Pierre Colombo. 2025. ColPali: Efficient document retrieval with vision language models. In *Proc. ICLR 2025*.

Nigel Fernandez, Alexander Scarlatos, and Andrew Lan. 2024. SyllabusQA: A course logistics question answering dataset. In *Proc. ACL 2024 (Long Papers)*, pages 10344–10369.

Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. Enabling large language models to generate text with citations. In *Proc. EMNLP 2023*, pages 6465–6488.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, pages 4878–4887.

Soyeong Jeong, Jinheon Baek, Sukmin Cho, Sung Ju Hwang, and Jong Park. 2024. Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity. In *Proc. NAACL 2024 (Long Papers)*.

Hailey Joren, Jianyi Zhang, Chun-Sung Ferng, Da-Cheng Juan, Ankur Taly, and Cyrus Rashtchian. 2025. Sufficient context: A new lens on retrieval augmented generation systems. In *Proc. ICLR 2025*.

Amita Kamath, Robin Jia, and Percy Liang. 2020. Selective question answering under domain shift. In *Proc. ACL 2020*, pages 5684–5696.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, pages 9459–9474.

A. Loison, Q. Macé, A. Edy, V. Xing, T. Balough, G. de Souza P. Moreira, B. Liu, M. Faysse, C. Hudelot, and G. Viaud. 2026. ViDoRe V3: A comprehensive evaluation of retrieval augmented generation in complex real-world scenarios. In *Proc. ACL 2026 (Long Papers)*, pages 16570–16600.

Yifei Ming, Senthil Purushwalkam, Shrey Pandit, Zixuan Ke, Xuan-Phi Nguyen, Caiming Xiong, and Shafiq Joty. 2025. FaithEval: Can your language model stay faithful to context, even if "the moon is made of marshmallows". In *Proc. ICLR 2025*.

Viktor Moskvoretskii, Maria Marina, Mikhail Salnikov, Nikolay Ivanov, Sergey Pletenev, Daria Galimzianova, Nikita Krayko, Vasily Konovalov, Irina Nikishina, and Alexander Panchenko. 2025. Adaptive retrieval without self-knowledge? bringing uncertainty back home. In *Proc. ACL 2025 (Long Papers)*.

Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, Kashun Shum, Randy Zhong, Juntong Song, and Tong Zhang. 2024. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In *Proc. ACL 2024 (Long Papers)*.

Anna Ostrowska, Michał Kukla, Gabriela Majstrak, Jan Opala, Sebastian Pergała, Jan Skwarek, and Anna Wróblewska. 2026. From surface learning to deep understanding: A grounded AI tutoring system for Moodle. In *Proc. IJCAI-26, Demonstrations Track*, pages 8479–8482.

Pranav Rajpurkar, Robin Jia, and Percy Liang. 2018. Know what you don't know: Unanswerable questions for SQuAD. In *Proc. ACL 2018 (Short Papers)*, pages 784–789.

Aviv Slobodkin, Omer Goldman, Avi Caciularu, Ido Dagan, and Shauli Ravfogel. 2023. The curious case of hallucinatory (un)answerability: Finding truths in the hidden states of over-confident large language models. In *Proc. EMNLP 2023*, pages 3607–3625.

Matyáš Veselý, Michal Průšek, and Jiří Franc. 2026. Your retriever already knows: Predicting RAG retrieval sufficiency from score distributions. In *Text, Speech, and Dialogue (TSD 2026), LNCS 16940*, pages 592–604. Springer.

A Week-by-week plan

Table A1 assigns every week to named owners (MV: Matyáš Veselý; GK: Grzegorz Kacprowicz; WK: Wiktoria Koniecko); course deadlines are set in bold.

Table A1: Week-by-week plan with named owners. Course deadlines in bold.

Week	Tasks	Owner
W1 4–10 Nov	Obtain the course materials or switch to the fallback dataset; set up the repository and the retrieval pipeline; draft the labelling guideline	MV pipeline; GK, WK guideline
W2 11–17 Nov	First baseline running end to end (proof of concept); first annotated questions; prepare the mid-term presentation and video	MV baseline; GK, WK questions
18 Nov	**Mid-term: 10-minute video and presentation** (state of the art and proof of concept)	all
W3 18–24 Nov	Continue annotation; implement the remaining baselines; set up the evaluation pipeline	GK, WK questions; MV, WK code
W4 25 Nov–1 Dec	Complete the question set and check annotator agreement; first comparative results; write the Phase 2 report and prepare the code and rebuttals	GK, WK data; MV results; all
2 Dec	**Phase 2: report, source code and rebuttals**	all
W5 2–8 Dec	Implement the gated variants; calibrate thresholds on the validation split	MV gates; WK thresholds
W6 9–15 Dec	Run the full experiments on the test split, including the missing-source stress test	MV, WK
W7 16–22 Dec	Human check of a sample of answers; error analysis	GK, WK check; all
23 Dec–5 Jan	Buffer (holiday break)	—
W8 6–12 Jan	Final results; write the final report; prepare the final presentation and video	all
13 Jan 2027	**Phase 3: final presentation, video and report**	all

B Contribution

Criterion	Contributor	Time
1.1 State of the art	Matyáš Veselý	5 h
1.2 Datasets and their status	Grzegorz Kacprowicz	5 h
1.3 Solution concept and experimental plan	Wiktoria Koniecko	4 h
1.4 Text quality and references	All three, jointly	1 h
1.5 Data statement	All three, jointly	0,5 h

Tasks and hours are consistent with the repository history.

C Declaration of LLM use

Large language model assistants (Anthropic Claude, OpenAI ChatGPT) were used to check the bibliographic details and numerical values in this document against the cited sources, to improve English wording and to suggest structure. The selection and reading of the literature, the research question, the pipeline and experimental design, the dataset design and all claims in this text are the authors' own; every value in Table 1 was verified by the authors against the published paper. No model-generated text was included without author review.